

Mini Compiler Design and Implementation:

Phase 1 - Lexical Specification and Tokenization

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Question 1: Lexical Specification and Tokenization (5 Marks)

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Using the given language specification (int, float, control constructs, operators, etc.), formally define the lexical structure of the language.

You are required to:

- Identify and classify all token categories relevant to the prescribed constructs.
- Specify the regular expressions corresponding to each token class.
- Implement a lexical analyzer that reads a source program and generates a token stream.
- Demonstrate the token stream generated for the prescribed evaluation program.
- Clearly report lexical errors, if any.

Your lexical analyzer must correctly tokenize the entire evaluation program provided.

1. Introduction

This project involves the design and implementation of a Lexical Analyzer (Scanner) for a mini-compiler language. The analyzer transforms raw source code into a structured stream of tokens, identifying keywords, identifiers, literals, and operators while flagging lexical errors.

2. Formal Lexical Specification

The following table defines the language's lexical structure using Regular Expressions (RE).

<u>Token Category</u>	<u>Description</u>	<u>Regular Expression</u>	<u>Example</u>
KEYWORD	Reserved control words	if else while print	while, if

DATATYPE	Primitive types	int float	int, float
IDENTIFIER	User-defined names	[a-zA-Z][a-zA-Z0-9_]*	myVar, sum
INTEGER_LITERAL	Whole numbers	0 [1-9][0-9]*	10, 420
FLOAT_LITERAL	Decimal numbers	[0-9]+\.[0-9]+	3.14, .5, 10.
RELATIONAL_OP	Comparison operators	== != <= >= < >	==, <
ARITHMETIC_OP	Math operators	+ - * / %	+, %
ASSIGNMENT_OP	Value assignment	=	=
DELIMITER	Statement separators	; ,	;
PARENTHESES	Grouping symbols	()	(,)
BRACES	Block delimiters	{ }	{, }
UNKNOWN	Unrecognized symbols	Any other character	@, \$

3. Implementation (C Source Code)

compi.l

```
%{
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "y.tab.h"

int line = 1;
%}

DIGIT      [0-9]
LETTER     [A-Za-z_]
ID         {LETTER}({LETTER}|{DIGIT})*
ICONST     0|[1-9]{DIGIT}*
FCONST     {DIGIT}+"."{DIGIT}+

%%

"int"      { return INT; }
"float"    { return FLOAT; }
"if"       { return IF; }
"else"     { return ELSE; }
"while"    { return WHILE; }
"print"    { return PRINT; }

{FCONST}   { yylval.fval = atof(yytext); return FCONST; }
{ICONST}   { yylval.ival = atoi(yytext); return ICONST; }

{ID}       { yylval.sval = strdup(yytext); return ID; }

"=="       { return EQ; }
"!="       { return NE; }
"<="       { return LE; }
">="       { return GE; }
"&&"       { return AND; }
"||"       { return OR; }

"<"        { return '<'; }
">"        { return '>'; }
"="        { return '='; }
```

```
"+"      { return '+'; }
"_"      { return '-'; }
"*"      { return '*'; }
"/"      { return '/'; }
"%"      { return '%'; }
"!"      { return '!'; }

"("      { return '('; }
")"      { return ')'; }
"{"      { return '{'; }
"}"      { return '}'; }
";"      { return ';'; }
","      { return ','; }

[ \t]+   { }
\n       { line++; }

.        { printf("Lexical error at line %d: %s\n", line, yytext); }

%%

int yywrap() { return 1; }
```

Compi.y

```
%{
#include <stdio.h>
#include <stdlib.h>

extern int line;
int yylex(void);
void yyerror(char *s);
%}

%union {
    int ival;
    float fval;
    char* sval;
}

%token <sval> ID
%token <ival> ICONST
%token <fval> FCONST
%token INT FLOAT IF ELSE WHILE PRINT
%token EQ NE LE GE AND OR

%%

program:
    program token
    | /* empty */
    ;

token:
    INT      { printf("Line %d: Found [int]\n", line); }
    | FLOAT  { printf("Line %d: Found [float]\n", line); }
    | IF      { printf("Line %d: Found [if]\n", line); }
    | ELSE    { printf("Line %d: Found [else]\n", line); }
    | WHILE   { printf("Line %d: Found [while]\n", line); }
    | PRINT   { printf("Line %d: Found [print]\n", line); }
    | ID      { printf("Line %d: Found ID [%s]\n", line, $1); free($1); }
    | ICONST  { printf("Line %d: Found Integer [%d]\n", line, $1); }
    | FCONST  { printf("Line %d: Found Float [%.2f]\n", line, $1); }
    | EQ      { printf("Line %d: Found Operator [==]\n", line); }
    | NE      { printf("Line %d: Found Operator [!=]\n", line); }
    | LE      { printf("Line %d: Found Operator [<=]\n", line); }
    | GE      { printf("Line %d: Found Operator [>=]\n", line); }
    | AND     { printf("Line %d: Found Operator [&&]\n", line); }
    | OR      { printf("Line %d: Found Operator [||]\n", line); }
    | '='     { printf("Line %d: Found Assignment [=]\n", line); }
```

```

| '+' { printf("Line %d: Found Math Op [+]\\n", line); }
| '-' { printf("Line %d: Found Math Op [-]\\n", line); }
| '*' { printf("Line %d: Found Math Op [*]\\n", line); }
| '/' { printf("Line %d: Found Math Op [/]\\n", line); }
| '%' { printf("Line %d: Found Math Op [%%]\\n", line); }
| '!' { printf("Line %d: Found Logic Op [!]\\n", line); }
| '<' { printf("Line %d: Found Comparison [<]\\n", line); }
| '>' { printf("Line %d: Found Comparison [>]\\n", line); }
| '(' { printf("Line %d: Found Paren [(]\\n", line); }
| ')' { printf("Line %d: Found Paren [)]\\n", line); }
| '{' { printf("Line %d: Found Brace [{]\\n", line); }
| '}' { printf("Line %d: Found Brace [}]\\n", line); }
| ';' { printf("Line %d: Found Semicolon [;]\\n", line); }
| ',' { printf("Line %d: Found Comma [,]\\n", line); }
;

%%

int main() {
    yyparse();
    return 0;
}

void yyerror(char *s) {
    fprintf(stderr, "Parse error at line %d: %s\\n", line, s);
}

```

4. Evaluation and Error Reporting

4.1 Input Source Code (Snippet)

```
int a;
int b;
int sum;
float avg;
a = 2 * (3 + 4);
b = 15;
sum = 0;
while (a < b && b != 0) {
    int temp;
    temp = a * 2;
    if ((temp % 3 == 0) || (a > 5)) {
        sum = sum + temp;
    } else {
        sum = sum - 1;
    }
    a = a + 1;
}
avg = sum / (b - a);
if (!(avg < 5.0)) {
    print(sum);
} else {
    print(avg);
}
```

4.2 Lexical Error Analysis

The sample input program provided in the assignment **did not contain any lexical errors** and was processed successfully by the scanner.

To verify the robustness of the lexical analyzer, the following **additional invalid symbols** were introduced separately for testing purposes:

1. **Illegal Character @**: The @ symbol is not defined in any regular expression for this language. The scanner flagged this as a LEXICAL_ERROR.
2. **Operators &**: The current specification defines individual relational and arithmetic operators. Characters like & were reported as unrecognized because they do not match the RELATIONAL_OP or ARITHMETIC_OP patterns.

In both cases, the scanner correctly identified and reported them as lexical errors. This confirms that the analyzer not only processes valid programs correctly but also handles invalid input reliably.

4.3 Final Token Stream

The following table represents the token stream generated by the lexical analyser from the input source code.

S.No	Token Type	Token Value
1	DATATYPE	int
2	IDENTIFIER	a
3	DELIMITER	;
4	DATATYPE	int
5	IDENTIFIER	b
6	DELIMITER	;
7	DATATYPE	int
8	IDENTIFIER	sum
9	DELIMITER	;
10	DATATYPE	float
11	IDENTIFIER	avg
12	DELIMITER	;
13	IDENTIFIER	a
14	ASSIGNMENT_OP	=
15	INTEGER_LITERAL	2
16	ARITHMETIC_OP	*
17	LPAREN	(
18	INTEGER_LITERAL	3
19	ARITHMETIC_OP	+

20	INTEGER_LITERAL	4
21	RPAREN	)
22	DELIMITER	;
23	IDENTIFIER	b
24	ASSIGNMENT_OP	=
25	INTEGER_LITERAL	15
26	DELIMITER	;
27	IDENTIFIER	sum
28	ASSIGNMENT_OP	=
29	INTEGER_LITERAL	0
30	DELIMITER	;
31	KEYWORD	while
32	LPAREN	(
33	IDENTIFIER	a
34	RELATIONAL_OP	<
35	IDENTIFIER	b
36	BOOLEAN_OP	&&
37	IDENTIFIER	b
38	RELATIONAL_OP	!=
39	INTEGER_LITERAL	0
40	RPAREN	)
41	LBACE	{
42	DATATYPE	int
43	IDENTIFIER	temp

44	DELIMITER	;
45	IDENTIFIER	temp
46	ASSIGNMENT_OP	=
47	IDENTIFIER	a
48	ARITHMETIC_OP	*
49	INTEGER_LITERAL	2
50	DELIMITER	;
51	KEYWORD	if
52	LPAREN	(
53	LPAREN	(
54	IDENTIFIER	temp
55	ARITHMETIC_OP	%
56	INTEGER_LITERAL	3
57	RELATIONAL_OP	==
58	INTEGER_LITERAL	0
59	RPAREN	)
60	BOOLEAN_OP	
61	LPAREN	(
62	IDENTIFIER	a
63	RELATIONAL_OP	>
64	INTEGER_LITERAL	5
65	RPAREN	)
66	RPAREN	)
67	LBACE	{

68	IDENTIFIER	sum
69	ASSIGNMENT_OP	=
70	IDENTIFIER	sum
71	ARITHMETIC_OP	+
72	IDENTIFIER	temp
73	DELIMITER	;
74	RBRACE	}
75	KEYWORD	else
76	LBRACE	{
77	IDENTIFIER	sum
78	ASSIGNMENT_OP	=
79	IDENTIFIER	sum
80	ARITHMETIC_OP	-
81	INTEGER_LITERAL	1
82	DELIMITER	;
83	RBRACE	}
84	IDENTIFIER	a
85	ASSIGNMENT_OP	=
86	IDENTIFIER	a
87	ARITHMETIC_OP	+
88	INTEGER_LITERAL	1
89	DELIMITER	;
90	RBRACE	}
91	IDENTIFIER	avg

92	ASSIGNMENT_OP	=
93	IDENTIFIER	sum
94	ARITHMETIC_OP	/
95	LPAREN	(
96	IDENTIFIER	b
97	ARITHMETIC_OP	-
98	IDENTIFIER	a
99	RPAREN	)
100	DELIMITER	;
101	KEYWORD	if
102	LPAREN	(
103	BOOLEAN_OP	!
104	LPAREN	(
105	IDENTIFIER	avg
106	RELATIONAL_OP	<
107	FLOAT_LITERAL	5.0
108	RPAREN	)
109	RPAREN	)
110	LBRACE	{
111	KEYWORD	print
112	LPAREN	(
113	IDENTIFIER	sum
114	RPAREN	)
115	DELIMITER	;

116	RBRACE	}
117	KEYWORD	else
118	LBRACE	{
119	KEYWORD	print
120	LPAREN	(
121	IDENTIFIER	avg
122	RPAREN	)
123	DELIMITER	;
124	RBRACE	}

Tokenization Summary:

- Total Tokens Generated: **124**
- Lexical Errors: **None**
- All constructs successfully recognized:
 - Data types
 - Identifiers
 - Arithmetic operators
 - Relational operators
 - Boolean operators
 - if-else statements
 - while loop
 - Block structures
 - print statement
 - Float literal (5.0)